

Steady State Library for Liquid Rocket Engine Cycle Design

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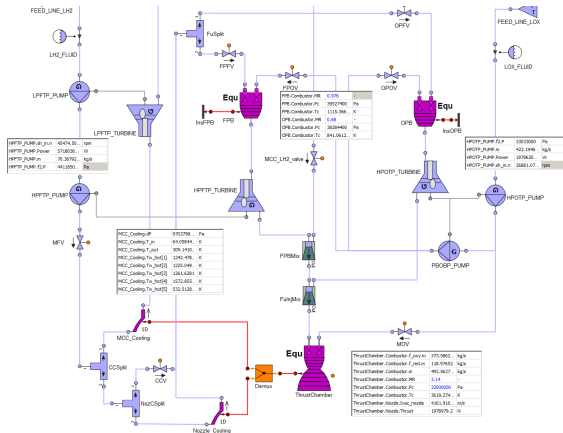
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New concepts for steady state design & analysis

Steady State Design:

- Need for a design tool for LRE cycles compatible with the transient libraries
- Parametric analysis
- Fast computation times
- Simplified models for preliminary designs
- More accurate models for off-design analyses



EcosimPro and the ESPSS library

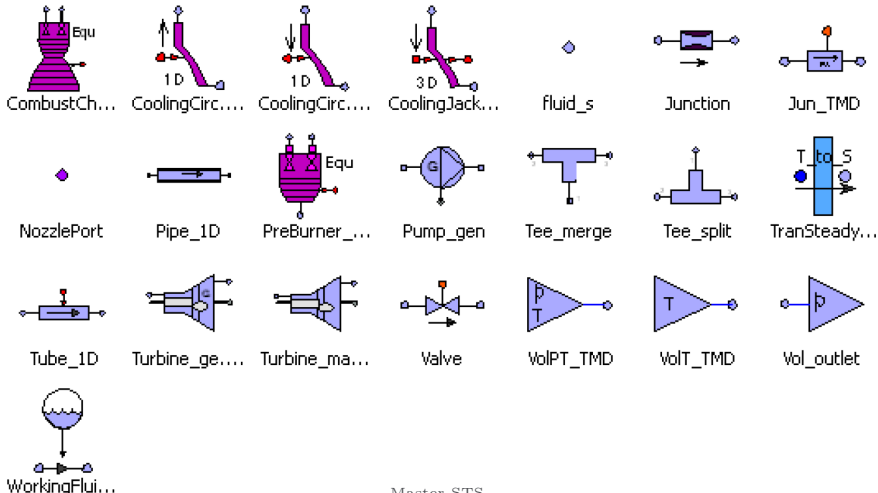
- The entire Steady State library has been implemented within the existing analysis software EcosimPro, designed to model various kinds of dynamic systems.
- ESPSS used to study only the transients of a propulsion system
- Steady State library considered as a completion of the ESPSS transient libraries
- Thermodynamic fluid properties from ESPSS
 - Detailed real gas models
 - Combustion properties by minimisation of Gibbs energy
 - Heat transfer and friction correlations

- Goal of this library is to have a tool for steady state design and off-design analyses of LRE cycles and has been validated using as comparison data of actual liquid rocket engine cycles

This goal is achieved by:

The EcosimPro Steady State library models, compatible with the ESPSS transient library

Steady State library : components overview



A new steady state port allows to propagate the main variables through the components

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-- (C) ESA-ESTEC, Propulsion & Aerothermodynamics Division
--
-- FILE NAME: Ports.el
-- LIBRARY: STEADY
-- DESCRIPTION: Definition of fluid port type
-- NOTES: few changes to propagate fluid composition
-- 15/06/2009 Version 0.9
-- AUTHORS: M. De Rosa, TEC-HPA, ESA-ESTEC.
-- F.Di Matteo, "Sapienza" University of Rome
-- VERIFIED: M. De Rosa
-- APPROVED:

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USE FLUID_PROPERTIES

PORT fluid

  --Flow variables (SUM type)
  SUM REAL m = 0 "Total mass flow (kg/s)"
  SUM IN REAL mh = 1e5 "Total enthalpy * flow (W)"

  --State variables (EQUAL type)
  EQUAL REAL P = 1e5 "Total pressure (Pa)"
  EQUAL OUT REAL h = 1e5 "Total enthalpy (J/kg)"

  --Combustion properties
  EQUAL OUT REAL fco2 = 0 "Auxiliar variable - set by combustor (d/kg)"
  EQUAL REAL N_eq[Chemicals] = 0 "Number of moles of chemical species"

  --Geometry, fluid (EQUAL type)
  EQUAL INTEGER n_fluid = 1 "Number of times the working fluid is defined (-)"
  EQUAL ENUM FluidKey fluid = PrGas_Air "Working fluid key"

  CONTINUOUS
  -- enthalpy flow
  mh = m * h

END PORT

```

The Steady State fluid port represents the basis and the rationale on which the whole library is coded:

- **SUM** variables will be summed at the ports
- **EQUAL** variables will be propagated to all components connected at the same port

Pipe 1D: Cylindrical 1D pipe with 1D heat transfer with the wall

- Mass conservation

$$\dot{m}_{in} = \dot{m}_{out} = \rho v A$$

- Momentum conservation

$$P_{i+1} = P_i - \Delta P_i$$

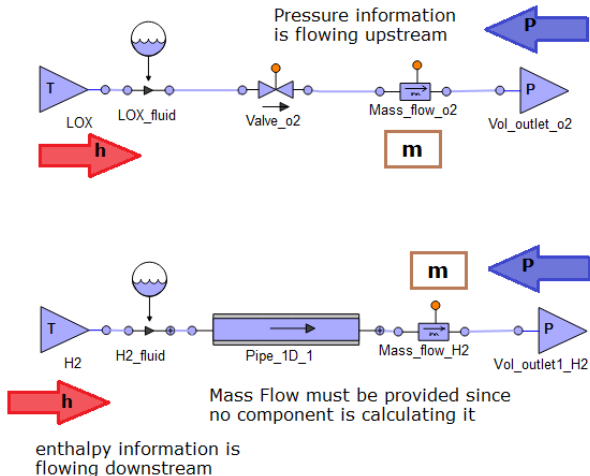
$$\Delta P = \frac{1}{2} \zeta \frac{\dot{m}^2}{\rho A^2}$$

- Energy conservation

$$\dot{m} \Delta H_i = \dot{q}_w \Delta A_{wet,i}$$

Most of the components calculates pressure drop. Only few of them provides mass flow and/or enthalpy information.

Special care must be taken once a system is created.



Cooling Jacket & Regenerative Circuit: 1D channels with 1D heat transfer with the walls

Cooling Jacket

- Inherited from the pipe component
- Mass flow, pressure drop and temperature rise of the coolant evaluated by using the same governing equations as for the pipes and the tubes
- Co-flow & counterflow directions modelled in two different components

Regenerative Circuit

- Simpler than the previous model
- Pre-defined pressure drop profile along the combustion chamber
- Pre-assigned hot gas side wall temperature $T_{w,hot}$ profile along the combustion chamber
- Channel width evaluated from the fixed number of channels
- Channel height input for the model
- Channel wall thickness is an output of the design

Combustor & Nozzle: 1D equilibrium combustion chamber

Design mode

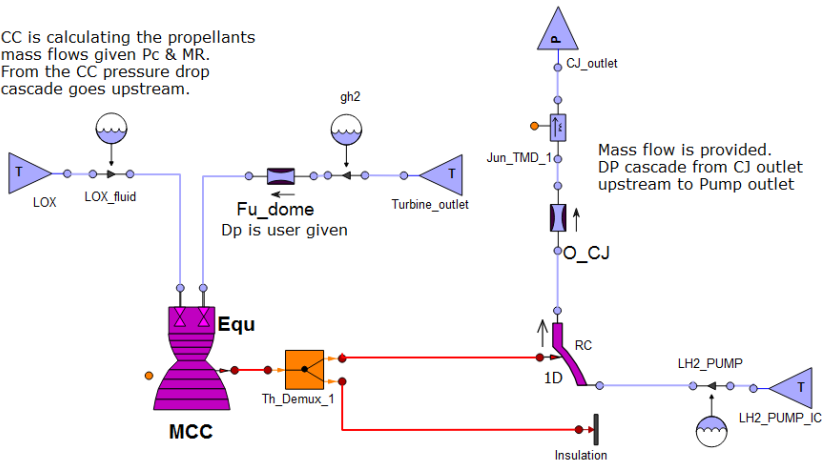
- Inputs: P_c , MR and Throat Area
- Outputs: Stagnation & Throat equilibrium composition, thermodynamic and transport properties; total mass flow
- Heat fluxes by Bartz correlation in closed form

Off Design mode

- Inputs: Throat Area, mass flows $\dot{m}_{ox}$, $\dot{m}_{fu}$
- Outputs: Stagnation & Throat equilibrium composition, thermodynamic and transport properties; chamber pressure P_c by iterative loop
- Heat fluxes by Bartz correlation in closed form

Combustion chamber is the main component of the system.
Once P_c & MR are given it calculates mass flows. Mass flow information propagates upstream to all components

CC is calculating the propellants mass flows given P_c & MR .
From the CC pressure drop cascade goes upstream.



Turbopump assembly: 0D components

Pump

The isentropic enthalpy rise is calculated assuming an isentropic transformation between inlet and outlet pressure.

- Design: Switch enabling user given N_s or rotational speed ω ; ΔP from ports
- Off Design: User given N_s , ω and ΔP are the outputs. Torque is an input from Turbine

Turbine

The isentropic enthalpy fall is calculated assuming an isentropic transformation between inlet and outlet pressure.

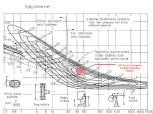
- Design: Switch for user given mass flow, pressure ratio Π , or pressure at ports
 - turbine_type = known_mflow. For closed cycles
 - turbine_type = known_pressures. For open cycles
 - turbine_type = known_PI_tt. For open cycles, especially in the pre-design phase
- Off Design: Rotational speed is an input from Pump, given mass flow

In each line the mass flow is given, to simulate information coming from TC component.

Each TP component uses a different switch (different working/behaving conditions)

$$D_3 = \frac{DH^{3/4}}{Q^{1/2}}$$

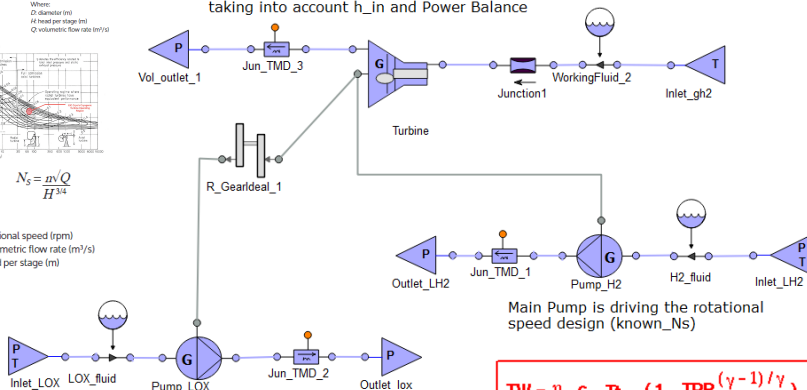
Where:
D: diameter (m)
H: head per stage (m)
Q: volumetric flow rate (m³/s)



$$N_s = \frac{n\sqrt{Q}}{H^{3/4}}$$

Where:
n: rotational speed (rpm)
Q: volumetric flow rate (m³/s)
H: head per stage (m)

Turbine is calculating Π given the mass flow and taking into account h_{in} and Power Balance



Secondary Pump is driven by the rotational speed design of the main pump (known_flow)

Main Pump is driving the rotational speed design (known_Ns)

$$TW = \eta_t c_p Tt_4 (1 - TPR^{(\gamma-1)/\gamma})$$

$$TPR = \frac{pt_5}{pt_4} = \left(\frac{Tt_5}{Tt_4} \right)^{\frac{\gamma}{\gamma-1}}$$

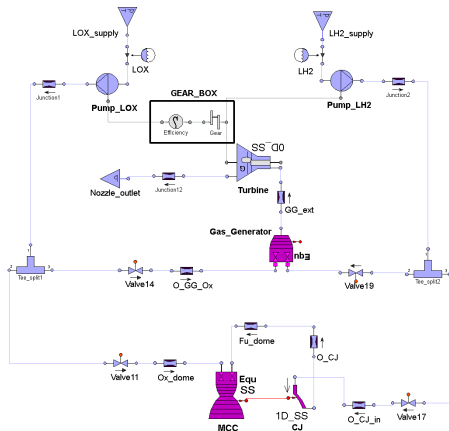
Validation Campaign:

- Qualitative/quantitative behaviour of each component validated (more than 40 test cases performed)
- Step by step approach to validate several components together (presented in previous works [?, ?]):
 - Pipeline network
 - Combustion Chamber model
 - HM7B Combustion chamber subsystem (Off-Design mode)
 - HM7B Turbo-Pump subsystem (Off-Design mode)
- Design of a real Gas Generator cycle: HM7B
- Design of a real Expander cycle: RL-10A-3-3A



- Propellants: Oxygen (O_2) - Hydrogen (H_2)
- Tank pressures & temperatures
- Chamber Pressure P_c
- Chamber Mixture Ratio MR_{cc}
- Chamber Throat Diameter D_t
- Gas Generator Mixture Ratio MR_{gg}
- Combustion efficiency η_{c*}
- Pump efficiencies η_p
- Pump specific speeds N_s
- Turbine efficiencies η_t

Design of a GG Cycle: HM7B - (1)



Main Outputs

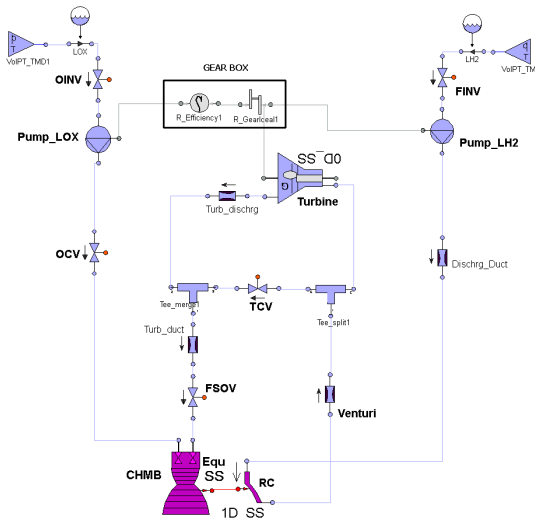
- Chamber Temperature T_c
- Mass flows $\dot{m}_{tot}$, $\dot{m}_{ox}$, $\dot{m}_{fu}$
- GG Pressure $P_{c,gg}$
- GG Mass flows $\dot{m}_{ox,gg}$, $\dot{m}_{fu,gg}$
- Thrust, Isp
- Heat flux $\dot{q}$
- Turbopump speeds ω
- Turbopump power W
- Turbine pressure ratio Π_{tt}
- CJ wall temperatures T_w
- CJ wall thickness t

Design of GG cycle: HM7B - (2)



Name	Nominal Value	Error	Name	Nominal Value	Error
$\dot{m}_{ox,gg}$ [kg/s]	14.86	0.59%	ω_t [rpm]	60500	0.16%
$\dot{m}_{fu,gg}$ [kg/s]		0.23%	τ_t [N·m]		4.48%
$\dot{m}_{t,gg}$ [kg/s]		0.3%	W_t [W]		4.32%
P_{gg} [bar]		0%	$T_{in,t}$ [K]		11.3%
T_{gg} [K]		1.03%	$\dot{m}_{p,LOX}$ [kg/s]		0.96%
$m_{cc,ox}$ [kg/s]		0.88%	ΔP_{LOX} [bar]	48	0.41%
$m_{cc,fu}$ [kg/s]		0.83%	$\omega_{p,ox}$ [rpm]	13000	0.16%
$m_{cc,t}$ [kg/s]		0.87%	$\tau_{p,ox}$ [N·m]		1.39%
m_{ch} [kg/s]		2.36%	$\dot{m}_{p,LH2}$ [kg/s]		7.56%
ΔP_{ch} [bar]		0.5%	ΔP_{LH2} [bar]	52	1.29%
$T_{out,ch}$ [K]		6.7%	$\omega_{p,fu}$ [rpm]	60500	0.16%
			$\tau_{p,fu}$ [N·m]		6.0%

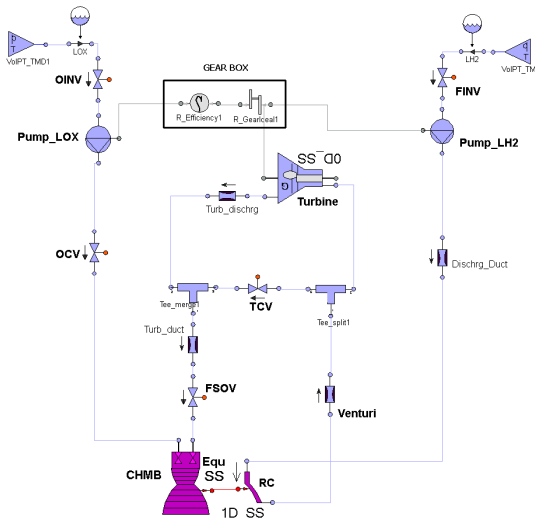
- Very good agreement with nominal data is recognisable
- Few parameters have an higher percentage error:
 - $\dot{m}_{p,LH2}$ does not take into account the dump and the tap-off mass flow rate vented from the engine
 - $T_{out,ch}$ takes into account the temperature increase in the injector dome. If compared to the LH2 injector dome temperature, the error decreases to 2.47%
 - Differences in turbomachinery components mainly due to $T_{in,t} <$ the nominal one



Design Requirements

- Propellants: Oxygen (O_2) - Hydrogen (H_2)
- Tank pressures & temperatures
- Chamber Pressure P_c
- Chamber Mixture Ratio MR
- Combustion efficiency η_{c*}
- Throat Diameter D_t
- Pump efficiencies η_p
- Pump specific speeds N_s
- Turbine efficiencies η_t

Design of Expander cycle: RL-10 - (1)



Main Outputs

- Chamber Temperature T_c
- Total mass flow $\dot{m}_{tot}$
- Ox mass flow $\dot{m}_{ox}$
- Fuel mass flow $\dot{m}_{fu}$
- Thrust, Isp
- Heat flux $\dot{q}$
- Turbopump speeds ω
- Turbopump power W
- Turbine pressure ratio Π_{tt}
- CJ wall temperatures T_w
- CJ wall thickness t

Design of Expander cycle: RL-10 - (2)

Name	Nominal Value []	Error
$\dot{m}_{cc,ox}$ [kg/s]	13.95	0.38%
$\dot{m}_{cc,fu}$ [kg/s]	2.76	0.38%
$\dot{m}_{cc,t}$ [kg/s]	16.71	0.38%
$\dot{m}_{ch}$ [kg/s]	2.76	0.38%
$T_{out,chu}$ [K]	213.44	0.18%
ω_t [rpm]	31 537	0.39%
$\dot{m}_t$ [kg/s]	2.69	1.1%
W_t [kW]	588.36	3.85%

Name	Nominal Value []	Error
$\dot{m}_{p,LOX}$ [kg/s]	13.95	0.38%
$\omega_{p,ox}$ [rpm]	12 948	0.4%
$W_{p,ox}$ [kW]	82.026	4.07%
$\dot{m}_{p,LH2}$ [kg/s]	2.7946	0.85%
$\omega_{p,LH2}$ [rpm]	31 537	0.39%
$W_{p,LH2}$ [kW]	501.86	3.68%
$Thrust$ [kN]	73.4	2.89%
Isp [s]	444	2.5%

- Very good agreement with nominal data: all errors < 5% [?, ?]
- $\dot{m}_{p,LH2}$ does not take into account the mass flow rate vented from the engine
- Turbomachinery power is regulated by the Thrust Control Valve (TCV) that is open at its nominal open area ratio of 9% [?]
- The RL-10 design model here described has proven to be very robust with respect to initial conditions: e.g. initial mass flow in the cooling jacket $\dot{m}_o$ is 2 kg/s, whereas the simulation result yields 2.76 kg/s

EcosimPro library for steady state applications has been presented

- System analysis approach
- Steady state design & off-design analysis for all LRE cycles
- Presence of a switch for “design/dimensioning” and “off-design/analysis” mode in each component
- Tailoring of system tools to specific needs via high level coding
- Design capability validated comparing simulation results with actual liquid rocket engines data

Any questions?

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